Software Engineering

Hand-in Assignment

Assignment 12

Instructions:

- Submit your solutions through GitHub Classroom and **remember** to submit a link to your repository in BrightSpace. Otherwise, I will have a hard time distinguishing between the different groups.
- When submitting, you should push your solutions to the existing repository. Follow the steps below as to what needs to be implemented in order to pass.
- You **must** add comments in your programs to explain your code. Any solutions submitted without any comments will not be graded.

Description

In today's lecture you were introduced to the concept of software reuse. The motivation for this is quite simple: the more functionality can be reused, the less code you will have to write from scratch. As a result the product can be shipped faster, or additional effort can be spent on developing a better product with more features.

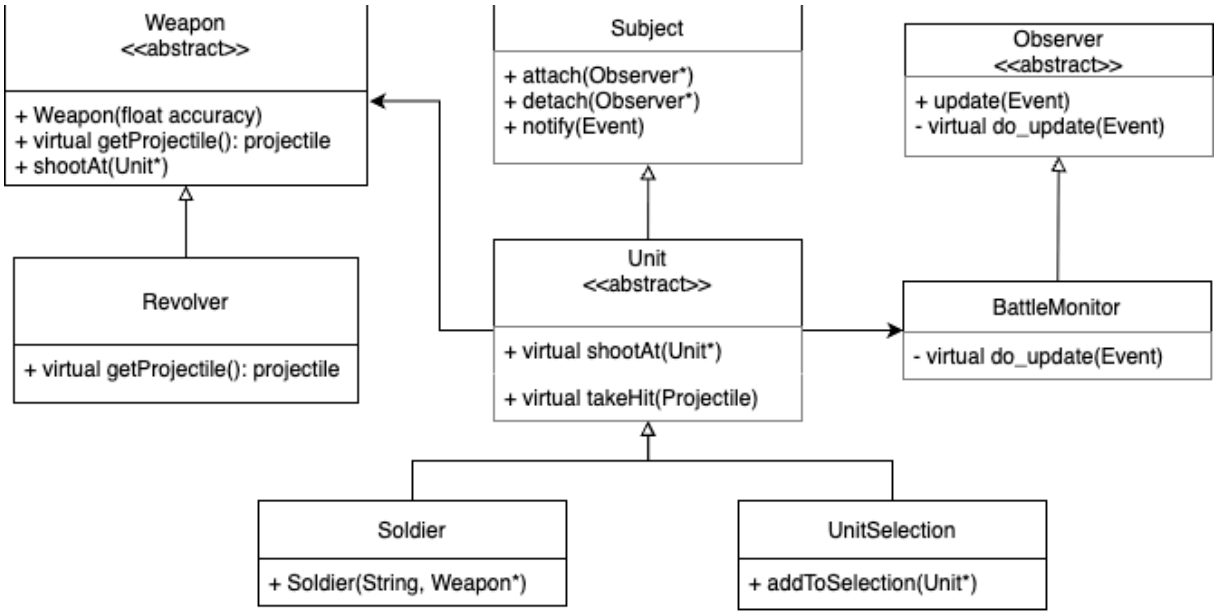
Code reuse happens at different levels. At the most general level, using standard libraries is a form of reuse. To see the amount of work you have saved, simply open the source files and you will quickly realize that you have saved quite a bit of work. At a higher level you may reuse common components for multiple products.

The key to writing reusable code is to realize that complex functionality can be implemented by combining several pieces of more generic functionality. In the context of object oriented programming, this means that we want to decompose complex classes into several smaller classes. The tools to achieve this are design patterns, which hopefully are a familiar concept by now.

Exercise

In this exercise we will implement part of a *real-time strategy* (RTS) game, like the Age of Empires, Warcraft, and Red Alert series. The structure is somewhat familiar to the one we saw in assignment 2.

Below is an overview of the classes:



- Unit: abstract class defining the interface of a unit.
- Weapon: abstract base class for weapons. The class implements the logic of shooting at units, and relies on the concrete implementation to produce the correct projectile.
- Subject: part of an observer pattern.
- Observer: part of observer pattern.
- UnitEvent: object inheriting from `Unit` and `Subject`, i.e. it is a unit which can also notify.
- **BattleMonitor**: an observer that prints information about which units have been hit.
- **Soldier**: concrete implementation of a unit that uses a weapon to shoot at other units.
- **UnitSelection**: composite pattern allowing multiple units to be treated as one.
- **Revolver**: concrete instance of weapon.

Classes in bold are to be implemented by you, the rest are provided.

Steps

Examine the test case described in `tests/src/test_wargame.cpp`. This should provide an overview of the intended functionality of the program.

1. Implement the `Revolver::get_projectile` method. This should return a reference to a projectile. *Hint: store the projectile as part of the object and return a reference to it.*
2. Implement the `Soldier::shootAt` method. This should invoke the `Weapon::shootAt` method.
3. Implement the `Soldier::takeHit` method. This should invoke the `Subject::notify` method with a string formatted as follows:

   ```
   ${name} got hit for ${projectile_damage} damage
   ```

   Hint: use the `stringstream` object to format the string.

4. Implement the `BattleMonitor::do_update` function, which is invoked whenever a unit is hit. This should store the message received through the event in the instance variable `messages`. You can access the message within the object like so:

```
UnitEvent *e = static_cast<UnitEvent *>(event);
std::cout << e->message << std::endl;
```

   The reason for the cast is related to templates and virtual methods and is not crucial to understand for this assignment.

5. Run the unit tests. Your code should now pass the first section of the tests.
6. Implement `UnitSelection::addToSelection`. This should store a pointer to the unit in the `selected` instance variable.
7. Implement `UnitSelection::shootAt`. This should cause each unit in the selection to attack the specified unit.
8. Implement `UnitSelection::takeHit`. This should cause each unit in the selection to take a hit, as if targeted normally.
9. Re-run the tests. Now the final test section should pass.